

Advantages

1. Much lower risk of electrical shock or shorting out due to moisture as there are no exposed conductors
2. Protected, corrosion-free connections as all the electronics is enclosed, away from water or oxygen in the atmosphere
3. Safer for medical implants—recharging/powering through the skin rather than wires penetrating the skin lowers the risk of infection
4. Convenience—the device can be placed on or close to a charge plate or stand rather than having to connect a power cable to it

portional to the separation distance between the plates (wireless power transfer through capacitive coupling explained later).

Laser. Laser is a device that emits light (electromagnetic radiation) through a process of optical amplification based on stimulated emission of photons. The term 'laser' is an acronym for 'Light Amplification by Stimulated Emission of Radiation.' The emitted laser light is notable for its high degree of spatial and temporal coherence.

Types of wireless electrical energy transfer

The aforementioned technologies work on different methods of electrical energy transfer without using a wired connection:

Electromagnetic induction. When magnetic flux flowing through a coil changes, an electromotive force (emf) is induced in the coil along with current. The famous Faraday's law (which relates to the induced emf in any closed loop including a closed circuit) is the central principle behind electromagnetic induction. Induction can be also be used as a means of wireless power transfer.

A changing current in one coil induces emf in another coil. The coils are not in contact and in this way energy can be very simply transported over short distances. The largest drawback of this method is that the short distance required for induction limits its application to very close-range situations.

Disadvantages

Lower efficiency, increased heat. Two main deterrents of inductive charging are its lower efficiency and increased resistive heating in comparison to the direct contact method. Implementation methods that use lower frequencies or older drive technologies charge much more slowly and generate more heat within most portable electronics, which is a big drawback.

Costly. Inductive charging requires drive electronics and coils in both the device and the charger, thus increasing the complexity and cost of manufacturing.

Slower charging. Due to the much lower efficiency, devices take longer to charge when the supplied power is equal.

Inconvenience. When a mobile device is connected to a cable for charging, it can still be freely moved around and operated. In most implementations of inductive charging (including the popular Qi standard), the mobile device must be left on the charging pad, which limits mobility.

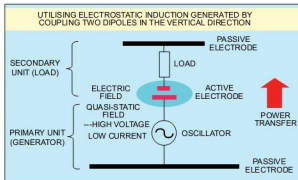


Fig. 3: Capacitive diagram

Electromagnetic radiation (microwave). Rectenna is the key component of wireless power transfer by radio waves. It is a combination of a rectifying circuit and an antenna. The antenna receives the electromagnetic power, which is converted into DC electric power by the rectifying circuit.

A simple rectenna can be constructed from a Schottky diode placed between the antenna dipoles. Schottky diodes are used because these have the lowest voltage drop and highest speed and therefore waste the least amount of power due to conduction and switching.

Since portable devices have small dimensions, the rectenna too should be small in size. A small antenna area results in a low amount of received power, which is a huge drawback. Because of these limitations, wireless power transfer using radio waves is mainly suitable for low-power applications like low-power wireless sensors.

Wave coupling. WiTricity (evanes-

cent wave coupling) is a technique that has recently been investigated by MIT researchers.

It basically extends the principle of magnetic induction to mid-range applications up to a few metres. The main differentiator is the use of resonance.

If the sender and the receiver have the same magnetic resonant frequency, energy can be efficiently transported while losses to the non-resonant environment are extremely small in comparison. Using resonance, for the same geometry, power can be transported approximately a hundred times more efficiently than without resonance.

One of the benefits of WiTricity is that since most common materials don't interact with magnetic fields, obstructing objects do not have much influence on the passage of WiTricity. The same goes for human tissue as well and hence health risks are low. The coils used for WiTricity are too large to be used in a cell phone, but the receiving coil can be reduced in size. According to the researchers at MIT, the transmitted power can be kept constant if the size of the sending coil is increased to keep the product of the sizes of both coils equal.

Electrostatic induction (capacitive coupling). In capacitive interface, the field is confined between conductive